Fiscal Year:	FY 2012	**Task Last Updated:**	FY 07/19/2012
PI Name:	Li, Chuan-Yuan Ph.D.		
Project Title:	A mechanistic investigation of space radiation-induced carcinogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27710	**Congressional District:**	4
Comments:	PI moved to Duke University in December 2011. Formerly at University of Colorado Denver (Ed., 2/8/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	06/01/2009	**End Date:**	03/31/2012
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 3/31/2012 per NSSC/A. Chu/ARC (Ed., 3/23/2011) NOTE: Received NCE until 5/31/2011 (previous end date 5/31/2010), per A. Chu/ARC (6/2010)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Li, Fang (Duke University)		
Grant/Contract No.:	NNX09AH19G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	One of major concerns for manned space missions of NASA is exposure to galactic cosmic rays (GCRs) or highly charged energetic (HZE) particles, which carries distinct health risks. The major goal of the NASA Bioastroanautics Roadmap and NASA ground-based studies in radiation biology is to assess potential risks of human exposure to HZE particles and to generate knowledge that can be used to mitigate the health risks of HZE particle exposure eventually. In this project, we will two specific aims to study HZE particle-induced mutagenesis and carcinogenesis in mammalian cells. These are: 1. To determine the potential interactions of reactive oxygen/nitrogen species and apoptosis in regulating HZE radiation-induced mutagenesis in mammalian cells. 2) to determine the roles of program cell death in HZE radiation induced mutagenesis/carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will have the following potential benefit for life on Earth: 1) we have developed a novel non-invasive technique to monitor radiation induced DNA damage though imaging H2AX foci, which has facilitated a better understanding radiation induced carcinogenesis in humans. 2) Our new technique have led to fundamental insights into how cells deal with radiation induced DNA damage. 3) We have also discovered a novel pathway called the "Phoenix Rising" where dying cells in mammalian tissue stimulates cellular proliferation. We showed that this pathway is important for both normal tissue regeneration and wound healing. It is also important for tumor cell re-population during radiotherapy. Both of these functions are counter-intuitive but very important. 4) We have also discovered novel functions for apoptotic caspases in epigenetic reprogramming during the induction of induced pluripotent stem cells. This unexpected function may facilitate better understand the roles of epigenetic reprogramming in radiation induced or other forms of carcinogenesis. 5). We have also developed a novel approach to reprogram differentiated primary human fibroblasts into dopaminergic neurons. This method should be useful for understanding Parkinson's disease and may provide a source for cell replacement therapy for Parkinson's disease.
Task Progress:	The major goals of this project are to understand the roles of free radicals and programmed cell death in space radiation induced mutagenesis and carcinogenesis. Towards these goals, we have made major progress in the funding period. Major achievements include: A). The development of a non-invasive bioluminescence-based imaging method to monitor gamma H2AX foci, a major indicator of DNA double strand breaks (DSBs) (Li et al, Cancer Research, 2011, 71:4130-37). Using this reporter, we showed that space radiation induced DSBs came in two waves. The first wave occurs within minutes and lasted for hours while the second wave occurs after 24 hrs and lasted for more than a week. The second wave of DNA damage has strong implications for space radiation induced DNA damage and carcinogenesis. This type of damage could be responsible for persistent genetic instability often observed after radiation exposure. B). The discovery of the “Phoenix Rising” pathway for wound healing and tissue regeneration. We discovered that dying cells in damaged tissues play a key role in mediating wound healing and tissue regeneration (Li et al, Science Signaling, 2010, 3:110: ra3) Surprisingly, caspase 3 and caspase 7, which are usually recognized as executioners of damaged or unwanted cells, play key facilitative roles regulating growth-promoting signals from dying cells. The elucidation of the counter-intuitive roles of apoptotic cells and apoptotic caspases significantly advanced our understanding of tissue homeostasis mechanisms in metazoan organisms. C). We also found that the same “Phoenix Rising” pathway was hijacked by tumors during radiotherapy. Dying cells in tumors exposed to radiation release potent caspase 3-controlled growth signals that promote the growth of surviving tumor cells, which fuels the repopulation of damaged tumor. Most importantly, consistent with these findings, higher levels of activated caspase 3 in tumor samples from human patients correlated with worse prognosis. These results are counter-intuitive and significantly changed the way we view cell death during tumor radiotherapy. D). Another discovery that we made concerns with surprising roles of caspases 3&8 in epigenetic reprogramming. We show that during induction of induced pluripotent stem (iPS) cells from human and mouse fibroblasts, caspases 3 and 8 are clearly activated. However, contrary to conventional wisdom, we found that blocking caspase activation did not increase the efficiency of iPSC induction. Instead, it significantly attenuated or completely blocked iPSC induction (Li et al, Cell Stem Cell, 2010, 7:508-20). These findings suggest that caspases played a facilitative role for epigenetic reprogramming, a role that has not been suggested before. Our paper was the cover story for Cell Stem Cell. This finding may have significant implications for space radiation and other forms of carcinogenesis. E). Still another significant finding in the funding period of this project involves the successful reprogramming of primary human fibroblasts into dopaminergic neurons (Liu et al, Cell Research, 22:321-332). We showed that a combination of 5 transcription factors (Mash1, Ngn2, Sox2, Nurr1, and Pitx3) can directly reprogram human primary fibroblasts into doparminergic neurons. These cells should stimulate research in providing a promising autologous source for cell replacement therapy for Parkinson’s disease. (Editor's Note: Project continues with "A mechanistic investigation of space radiation-induced carcinogenesis--NNX12AB88G," resulting from PI move to Duke University in December 2011. See that record for continuing reports.)
Bibliography Type:	Description: (Last Updated: 10/30/2019)
Articles in Peer-reviewed Journals	Liu X, Li F, Stubblefield EA, Blanchard B, Richards TL, Larson GA, He Y, Huang Q, Tan AC, Zhang D, Benke TA, Sladek JR, Zahniser NR, Li CY. "Direct reprogrammming of human fibroblasts into dopaminergic neuron-like cells." Cell Research. 2012 Feb;22(2):321-32. Epub 2011 Nov 22. PubMed PMID: 22105488 ; http://dx.doi.org/10.1038/cr.2011.181 , Feb-2012

Articles in Peer-reviewed Journals	Brogan J, Li F, Li W, He Z, Huang Q, Li CY. "Imaging molecular pathways: Reporter genes." Radiation Research. 2012 Apr;177(4):508-13. Epub 2012 Feb 21. Review. PMID: 22348248 , Apr-2012
Articles in Peer-reviewed Journals	Kon T, Zhang X, Huang Q, Yang Z, Liu S, Yan, B, Li F, Wang H, Li CY. "Oncolytic virus-mediated tumor radiosensitization in mice through DNA-PKcs-specific shRNA." Translational Cancer Research 2012 Jun;1(1):6-14. See http://www.thetcr.org/issue/view/36/showToc , Jun-2012
Articles in Peer-reviewed Journals	Huang Q, Li F, Liu X, Li W, Shi W, Liu FF, O'Sullivan B, He Z, Peng Y, Tan AC, Zhou L, Shen J, Han G, Wang XJ, Thorburn J, Thorburn A, Jimeno A, Raben D, Bedford JS, Li CY. "Caspase 3-mediated stimulation of tumor cell repopulation during cancer radiotherapy." Nature Medicine. 2011 Jul 3;17(7):860-6. PubMed PMID: 21725296 ; http://dx.doi.org/10.1038/nm.2385 , Jul-2011